Business Requirements Document (BRD)

AI Early-Warning System for Accounts Receivable

AI Builder Cup – Singapore | Version 0.2 — English Draft

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Document Purpose

This Business Requirements Document defines the business needs, objectives, processes, functional requirements, data requirements, and success criteria for developing an AI-assisted early-warning system to identify potential customer payment delays during active truck rental contracts.

The document follows a structured Business Analysis approach:

Business Problem → Management Decisions → KPIs → Business Processes → Functional Requirements → Data Requirements → Success Criteria

To maintain analytical integrity, information is classified as follows:

- **[Defined]** Requirements and design decisions established during previous discussions.
- **[Assumption]** Proposed conditions, thresholds, or design choices requiring validation.

- **[TBD]** Information requiring further clarification or stakeholder approval.

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1. Executive Summary

1.1 Business Background

Truck rental and leasing companies are exposed to financial risks associated with delayed customer payments throughout the contract lifecycle.

Credit assessments are generally conducted before a rental contract is approved. However, customers' financial conditions and payment behavior may change significantly during an active contract.

Without continuous risk monitoring, these changes may remain undetected until invoices become overdue.

Consequently, the company may experience increasing overdue accounts receivable, delayed cash inflows, higher collection costs, and potentially truck repossession.

1.2 Business Problem Statement

No Early-Warning Monitoring

> The company lacks a continuous, proactive monitoring mechanism to identify deteriorating customer payment behavior during active truck rental contracts, resulting in delayed risk detection and reactive collection activities.

The fundamental problem is not simply that customers fail to pay on time.

The underlying issue is the company's limited ability to identify increasing payment risks before invoices become overdue.

1.3 Proposed Business Solution

The proposed **AI Early-Warning System for Accounts Receivable** will leverage existing financial and operational data to:

1. Continuously monitor customer payment behavior.
2. Identify deteriorating payment patterns and increasing financial exposure.
3. Classify customers into risk categories.
4. Generate alerts before invoice due dates.

5. Support preventive collection activities.
6. Measure the effectiveness of risk monitoring and intervention.

The Minimum Viable Product (MVP) will primarily use a **rule-based risk scoring mechanism**.

Gemini AI will support risk explanations and recommended actions. It will not independently approve credit decisions, terminate contracts, or initiate repossession.

1.4 Expected Business Benefits

Business Benefit	Expected Improvement
Risk Visibility	Continuous visibility into customer payment risks
Early Detection	Earlier identification of deteriorating payment behavior
Collection Efficiency	Risk-based prioritization of collection activities
Cash Flow Management	Reduced accumulation of overdue receivables
Management Decision Support	Better-informed contract and exposure decisions
Operational Accountability	Traceable alerts, responsibilities, and collection actions

[Assumption] These are expected benefits, not outcomes that have already been demonstrated.

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2. Business Objectives

2.1 Primary Business Objective

To improve the company's ability to identify and respond to potential customer payment delays before invoices become overdue.

2.2 Specific Business Objectives

ID	Business Objective	Measurement KPI
BO-01	Reduce the proportion of overdue receivables	K02 – Overdue AR Ratio
BO-02	Monitor customer payment reliability	K06 – On-Time Payment Rate
BO-03	Detect deterioration in payment behavior	K07 – Payment Delay Trend
BO-04	Identify increasing customer financial exposure	K10 – Exposure Growth Rate
BO-05	Ensure eligible active contracts are monitored	K15 – Monitoring Coverage
BO-06	Ensure timely responses to risk alerts	K17 – Alert Response SLA
BO-07	Measure the effectiveness of pre-due risk detection	K21 – Pre-Due Detection Rate
BO-08	Ensure alerts provide sufficient intervention time	K24 – Alert Lead Time

K22 – Alert Precision is designated as a mandatory MVP quality guardrail.

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3. Business Problem Analysis

3.1 Current Business Challenges

ID	Business Challenge	Potential Business Impact
BC-01	Credit assessment is concentrated at contract initiation	Changes in customer risk may go unnoticed
BC-02	Payment behavior is not proactively monitored	Deteriorating payment patterns may be detected too late
BC-03	Collection priorities are primarily based on overdue invoices	Collection activities become reactive
BC-04	Financial exposure is not consistently assessed alongside payment behavior	Increasing financial risk may be underestimated

ID	Business Challenge	Potential Business Impact
BC-05	Early-warning effectiveness is not systematically measured	Management cannot reliably assess detection performance
BC-06	Alert follow-up is not consistently structured	Delayed collection responses and unclear accountability

[Assumption] These challenges represent hypotheses about the current operating environment and require validation through stakeholder interviews and process observation.

3.2 Root Cause Hypotheses

Category	Potential Root Cause
People	Collection teams primarily focus on already-overdue invoices
Process	No standardized continuous risk monitoring process
Data	Invoice, payment, contract, and exposure data are not consistently analyzed together
Technology	No dedicated pre-due risk engine or alert mechanism
Governance	Limited monitoring of alert effectiveness and response SLAs

3.3 Business Consequences

The proposed solution aims to interrupt the following risk escalation:

Payment Behavior Deterioration → Undetected Risk → Overdue Invoice → Collection Escalation → Potential Repossession

Repossession represents a possible severe outcome, not an automatic consequence of overdue payments.

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4. Stakeholder Analysis

Stakeholder	Business Responsibility	Expected System Support
Finance / Accounts Receivable	Monitor receivables and overdue balances	AR monitoring and KPI dashboards
Collection Officer	Contact customers and follow up on payments	Prioritized alerts and action tracking
Collection Manager	Manage collection priorities and escalations	Risk dashboards and SLA monitoring
Credit / Risk Team	Evaluate customer payment risks	Payment behavior analysis and risk classification
Credit Manager	Review customer exposure and credit decisions	Exposure monitoring and risk history
Sales / Account Management	Manage customer relationships and contract renewals	Risk visibility before contract renewal
Fleet / Operations	Monitor vehicle operations	Vehicle utilization insights in future phases
IT / Data Team	Maintain data integration and system reliability	Data validation and processing monitoring
Senior Management	Oversee financial risk and business performance	Portfolio-level KPIs and management dashboards
Legal / Asset Recovery	Handle contractual enforcement and asset recovery	Contract and collection history

[TBD] Actual stakeholder assignments and the RACI matrix must be confirmed.

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5. Management Decision Requirements

The system must provide relevant information to support the following management decisions.

Decision ID	Management Decision	Required Business Information
D01	Continue Rental	Payment behavior, risk classification, current exposure
D02	Renew Rental Contract	Payment history, overdue history, risk trends
D03	Increase Deposit / Security	Financial exposure, security coverage, customer risk
D04	Prioritize Collection	Risk classification, invoice due dates, alert history
D05	Restructure Payment	Payment history, outstanding receivables, collection history
D06	Freeze Additional Units	Financial exposure, credit limits, payment risk
D07	Evaluate Repossession	Severe delinquency, contractual conditions, collection history

Management Decision Rule

The system provides risk information and decision support.

Commercial decisions, payment restructuring, restrictions on additional vehicle deployment, and repossession must remain subject to authorized human approval and applicable contractual requirements.

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6. Business Process Requirements

6.1 AS-IS Business Process

[Hypothesis requiring validation]

Credit Assessment → Contract Approval → Truck Deployment → Invoice Generation → Payment Due Date → Invoice Overdue → Collection Activities → Escalation / Recovery → Potential Repossession

Identified Process Gap

The current process is assumed to lack a sufficiently proactive monitoring mechanism between contract activation and invoice due dates.

As a result, collection intervention may begin only after payment deterioration has already resulted in overdue receivables.

6.2 TO-BE Business Process

Data Foundation → Behavioral Risk Monitoring → Risk Engine → Risk Classification → Pre-Due Alert Generation → Collection Action → KPI Monitoring → Management Review

The proposed operating model consists of five core functional groups.

Process Group	Key Activities	Expected Output
1. Data Foundation	Retrieve and validate contract, invoice, payment, and exposure data	Validated monitoring dataset
2. Behavioral Risk Monitoring	Calculate K06, K07, and K10	Customer risk indicators
3. Risk Engine	Evaluate indicators and classify customer risk	Risk score and classification
4. Alert Management	Generate alerts, assign owners, and record actions	Alert and collection action records
5. KPI Monitoring	Calculate and display system performance KPIs	Management dashboard

6.3 Business Process Mapping

BP ID	Business Process	Relevance to the Solution
BP01	Credit Assessment	Credit evaluation and contract approval decisions
BP02	Contract Management	Active contract information and financial exposure
BP03	Contract & Billing	Invoice generation, due dates, and payment terms
BP04	Accounts Receivable	Outstanding balances, payments, and reconciliation
BP05	Collection Management	Alert responses and collection activities

BP ID	Business Process	Relevance to the Solution
BP06	Risk Monitoring	Risk calculations, classifications, and alerts
BP07	Unit Monitoring	Vehicle utilization and operational information
BP08	Asset Recovery	Contractual escalation and recovery activities

MVP Focus: BP02–BP06, with BP01 consuming risk information for credit-related decisions. BP07 and BP08 are primarily associated with future integration or downstream business processes.

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7. KPI Framework

7.1 KPI Prioritization Principles

A total of **25 KPIs** have been identified across four categories:

1. Outcome KPIs
2. Leading Indicators
3. Process KPIs
4. Alert Quality KPIs

Eight KPIs have been selected as MVP priorities based on:

- Direct relevance to the early-warning business problem.
- Ability to support preventive decisions.
- Availability of existing internal data.
- Relative implementation complexity and cost.
- Ability to demonstrate measurable business value.

One additional KPI, **K22 – Alert Precision**, is mandatory as a quality guardrail.

7.2 Outcome KPIs

KPI ID	KPI Name	Formula	Green	Yellow	Red	MVP
K01	Days Sales Outstanding	Average AR / Net Credit Sales × Days	⩽35 days	>35–45 days	>45 days	No

KPI ID	KPI Name	Formula	Green	Yellow	Red	MVP
K02	**Overdue AR Ratio**	Overdue AR / Total AR × 100%	<10%	10–20%	>20%	**Yes**
K03	30+ DPD Ratio	AR with DPD >30 / Total AR × 100%	<5%	5–10%	>10%	No
K04	Bad Debt Write-off Ratio	Net Write-offs / Net Credit Sales × 100%	<1%	1–2%	>2%	No
K05	Repossession Rate	Repossessed Contracts / Exposed Contracts × 100%	<1%	1–3%	>3%	No

[Assumption] The DSO thresholds are based on a Net 30 payment-term scenario. The denominator for the repossession rate must be consistently defined for the measurement period.

7.3 Leading Indicator KPIs

KPI ID	KPI Name	Formula	Green	Yellow	Red	MVP
K06	**On-Time Payment Rate**	On-Time Settled Invoices / Total Settled Invoices × 100%	≥95%	80–<95%	<80%	**Yes**
K07	**Payment Delay Trend**	Average Delay of Last 3 − Average Delay of Previous 3	≤2 days	>2–5 days	>5 days	**Yes**
K08	Partial Payment Frequency	Partially Paid Invoices / Relevant Paid Invoices × 100%	<10%	10–25%	>25%	No
K09	Promise-to-Pay Breach Rate	Broken Promises / Matured Promises × 100%	<10%	10–25%	>25%	No
K10	**Exposure Growth Rate**	(Current Exposure − Previous Exposure) / Previous Exposure × 100%	≤10%	>10–20%	>20%	**Yes**
K11	Credit Limit Utilization	Current Exposure / Approved Credit Limit × 100%	<70%	70–90%	>90%	No
K12	Billing Dispute Rate	Disputed Amount / Open AR × 100%	<5%	5–10%	>10%	No
K13	Unit Utilization Decline	(Baseline Usage − Current Usage) / Baseline Usage × 100%	<15%	15–30%	>30%	No
K14	Security Coverage Ratio	Eligible Security / Current Exposure × 100%	≥100%	75–<100%	<75%	No

Calculation Rules

K07 requires at least six historically settled invoices. If sufficient history is unavailable, the system must return **Insufficient Data**, rather than automatically assigning a low-risk classification.

K10 measures exposure growth over a proposed 30-day period.

Zero denominators, incomplete payment allocations, and missing historical values must be handled explicitly.

7.4 Process KPIs

KPI ID	KPI Name	Formula	Green	Yellow	Red	MVP
K15	**Monitoring Coverage**	Assessed Eligible Contracts / Total Eligible Contracts × 100%	≥95%	85–<95%	<85%	**Yes**
K16	Risk Detection Latency	Alert Timestamp − Signal Availability Timestamp	≤24 hours	>24–48 hours	>48 hours	No
K17	**Alert Response SLA Compliance**	Alerts Responded Within SLA / Actionable Alerts × 100%	≥90%	75–<90%	<75%	**Yes**
K18	Preventive Action Completion	Actions Completed on Time / Due Actions × 100%	≥90%	75–<90%	<75%	No
K19	Reconciliation Accuracy	Correct Payment Matches / Received Payments × 100%	≥99%	95–<99%	<95%	No
K20	High-Risk Review Coverage	Reviewed High-Risk Customers / Total High-Risk Customers × 100%	≥95%	80–<95%	<80%	No

[Assumption] K17 uses a maximum response time of one business day for the first meaningful collection action.

7.5 Alert Quality KPIs

KPI ID	KPI Name	Formula	Green	Yellow	Red	MVP
K21	**Pre-Due Detection Rate / Recall**	TP / (TP + FN) × 100%	≥60%	40–<60%	<40%	**Yes**
K22	Alert Precision	TP / (TP + FP) × 100%	≥50%	30–<50%	<30%	**Guardrail**
K23	False Positive Rate	FP / (FP + TN) × 100%	<10%	10–20%	>20%	No

KPI ID	KPI Name	Formula	Green	Yellow	Red	MVP
K24	**Alert Lead Time**	Due Date − First Pre-Due Alert Date	⩾5 days	2–<5 days	<2 days	**Yes**
K25	Alert-to-Action Conversion	Actioned Alerts / Actionable Alerts × 100%	⩾80%	60–<80%	<60%	No

[Assumption] All KPI thresholds are initial design proposals and must be calibrated against actual historical performance.

7.6 MVP KPI Success Targets

The following success targets are distinct from the KPI classification thresholds.

KPI	Proposed MVP Target	Measurement Purpose
K02 Overdue AR Ratio	⩾10% relative reduction within 3–6 months	Business outcome
K06 On-Time Payment Rate	TBD	Customer payment behavior
K07 Payment Delay Trend	TBD	Behavioral deterioration
K10 Exposure Growth Rate	TBD	Financial exposure
K15 Monitoring Coverage	⩾95%	Monitoring completeness
K17 Alert Response SLA	⩾90% within one business day	Operational responsiveness
K21 Pre-Due Detection Rate	Recall ⩾60%	Detection effectiveness
K24 Alert Lead Time	⩾5 days	Preventive intervention window
K22 Alert Precision	⩾50% **[Assumption]**	Alert quality guardrail

Targets for K06, K07, and K10 remain **TBD** pending historical baseline analysis.

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8. Business Requirements

BR ID	Business Requirement	Business Rationale	Priority
BR-01	The system must monitor eligible customers with active rental contracts	Eliminate monitoring blind spots	Must
BR-02	The system must analyze historical customer payment behavior	Identify payment patterns	Must
BR-03	The system must detect deterioration in payment behavior	Enable early-warning capabilities	Must
BR-04	The system must monitor changes in customer financial exposure	Identify increasing financial risk	Must
BR-05	The system must classify customer risk levels	Support collection prioritization	Must
BR-06	The system must generate alerts before invoice due dates	Enable preventive intervention	Must
BR-07	The system must record alert ownership and collection responses	Establish operational accountability	Must
BR-08	The system must measure monitoring and alert effectiveness	Support performance evaluation	Must
BR-09	The system should provide explanations of risk indicators	Improve user understanding	Should
BR-10	The system should support contract and exposure reviews	Improve management decisions	Should
BR-11	The system may incorporate vehicle utilization information	Extend risk intelligence	Nice
BR-12	The system may integrate external credit information	Enhance future risk assessments	Nice

Priority Definitions: Must = essential for MVP; Should = important but may initially use a simplified implementation; Nice = future enhancement.

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9. Functional Requirements

9.1 Functional Requirement Input (FRI)

The following requirement identifiers are preserved from the previous analysis and Diagrams 6–11.

FR ID	Functional Requirement	Related KPI
FRI-01	Retrieve Invoice and Payment History	K06
FRI-02	Calculate On-Time Payment Rate	K06
FRI-03	Retrieve Historical Settlement Dates	K07
FRI-04	Calculate Payment Delay Trend	K07
FRI-05	Retrieve AR Outstanding Balance	K02
FRI-06	Calculate Overdue AR Ratio	K02
FRI-07	Retrieve Historical Exposure	K10
FRI-08	Calculate Exposure Growth Rate	K10
FRI-09	Retrieve Alert Timestamp	K17
FRI-10	Retrieve Collection Action Timestamp	K17
FRI-11	Calculate SLA Compliance	K17
FRI-12	Retrieve Alert and Due Dates	K21
FRI-13	Retrieve Actual Payment Outcome	K21
FRI-14	Calculate Detection Recall	K21
FRI-15	Retrieve First Pre-Due Alert Date	K24
FRI-16	Calculate Alert Lead Time	K24
FRI-17	Retrieve Eligible Active Contracts	K15
FRI-18	Retrieve Risk Assessment Records	K15
FRI-19	Calculate Monitoring Coverage	K15

9.2 Functional Requirement Output (FRO)

FR ID	Functional Requirement	Related KPI
FRO-01	Display Payment Performance	K06
FRO-02	Generate Payment Behavior Signal	K06
FRO-03	Display Payment Delay Trend	K07
FRO-04	Generate Deterioration Signal	K07
FRO-05	Display Overdue AR Ratio	K02
FRO-07	Display Exposure Growth Trend	K10
FRO-08	Generate Exposure Risk Signal	K10
FRO-09	Display Alert Response SLA	K17
FRO-10	Generate SLA Breach Notification	K17
FRO-11	Display Pre-Due Detection Rate	K21
FRO-13	Display Alert Lead Time	K24
FRO-15	Display Monitoring Coverage	K15
FRO-16	Generate Unmonitored Contract List	K15

Requirement Numbering Note

FRO-06, FRO-12, and FRO-14 are not defined in the available reference material.

These identifiers have intentionally not been reassigned to avoid inconsistencies with previously established diagrams.

9.3 Functional Requirement Gaps

The TO-BE process analysis identified six capabilities without dedicated functional requirement identifiers.

Gap ID	Functional Capability	Current Status
GAP-01	Daily Risk Assessment	No dedicated FR
GAP-02	Risk Score Calculation	No dedicated FR

Gap ID	Functional Capability	Current Status
GAP-03	Green / Yellow / Red Risk Classification	No dedicated FR
GAP-04	Alert Generation and Assignment	No dedicated FR
GAP-05	Collection Action Recording	No dedicated FR
GAP-06	Insufficient Data Handling	No dedicated FR

Recommendation: These capabilities should receive dedicated functional requirement identifiers before the BRD becomes the approved implementation baseline.

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10. Business Rules

Rule ID	Business Rule	Priority
RULE-01	Only eligible active contracts shall be included in monitoring	Must
RULE-02	Risk assessments shall run periodically, with daily execution as the initial target	Must
RULE-03	Risk scores shall be calculated using K06, K07, and K10	Must
RULE-04	Customers shall be classified as Green, Yellow, or Red	Must
RULE-05	Insufficient historical data shall produce an Insufficient Data status	Must
RULE-06	Every alert shall have a unique identifier and timestamp	Must
RULE-07	Every actionable alert shall have an assigned owner	Must
RULE-08	Collection responses shall be recorded with timestamps	Must
RULE-09	Duplicate alerts for the same risk condition shall be controlled	Must

Rule ID	Business Rule	Priority
RULE-10	Gemini shall not independently modify risk scores	Must
RULE-11	High-impact contractual decisions shall require human authorization	Must
RULE-12	Prediction calculations shall not use information unavailable at prediction time	Must

10.1 Proposed Risk Scoring Methodology

[Assumption] The MVP will use the following weighted scoring model.

Risk Indicator	Weight
K06 – On-Time Payment Rate	40%
K07 – Payment Delay Trend	40%
K10 – Exposure Growth Rate	20%
Total	**100%**

Each indicator receives a score based on its risk classification.

Indicator Status	Score
Green	0
Yellow	50
Red	100

Risk Score Formula

Risk Score = 0.4 × (K06 Score) + 0.4 × (K07 Score) + 0.2 × (K10 Score)

Risk Classification

Risk Score	Risk Classification
0–<30	Green

Risk Score	Risk Classification
30–<60	Yellow
60–100	Red

Important: This score is a rule-based risk prioritization index, not a calibrated probability of default.

Weights and thresholds must be validated against historical customer payment outcomes.

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11. Data Requirements

11.1 Logical Data Architecture

The MVP will prioritize existing internal data sources.

ERP / Finance / Contract Management → Data Integration → Data Store → Behavioral KPI Engine → Risk Engine → Alert Management → KPI Dashboard

[Assumption] Source system names represent functional system categories rather than verified applications used by a specific company.

11.2 Core Data Entities

Entity	Primary Key	Business Purpose
CUSTOMER	customer_id	Customer identification
CONTRACT	contract_id	Rental contract information
INVOICE	invoice_id	Billing information
PAYMENT_TRANSACTION	payment_id	Payment transactions
PAYMENT_ALLOCATION	allocation_id	Invoice-level payment allocation
AR_DAILY_SNAPSHOT	snapshot_id	Historical receivables balances
RISK_ASSESSMENT	assessment_id	Risk assessment records
ALERT	alert_id	Risk alert history
COLLECTION_ACTION	action_id	Collection follow-up records

11.3 Data Element Requirements

Data Element	Proposed Source System	Priority	Related KPI
Customer ID	ERP / Customer Master	Must	K06, K07, K10
Contract ID	ERP / Contract Management	Must	K10, K15
Contract Status	Contract Management	Must	K15
Contract Start / End Date	Contract Management	Must	K15
Invoice ID	ERP / Billing	Must	K02, K06, K07
Invoice Date	ERP / Billing	Must	K06, K07
Due Date	ERP / Billing	Must	K02, K06, K07, K24
Invoice Amount	ERP / Billing	Must	K02, K06
Payment Date	Finance	Must	K06, K07
Payment Amount	Finance	Must	K06, K07

Data Element	Proposed Source System	Priority	Related KPI
Payment Allocation	Finance	Must	K06, K07
Outstanding AR	Finance	Must	K02
Historical Exposure	Finance	Must	K10
Risk Assessment Date	Risk Engine	Must	K15
Risk Level	Risk Engine	Must	Alert Management
Alert ID	Risk Engine	Must	K17, K21, K24
Alert Timestamp	Risk Engine	Must	K17, K21, K24
Collection Action Timestamp	CRM / Collection	Must	K17
Actual Payment Outcome	Finance	Must	K21
Credit Limit	ERP / Credit	Should	K11
Security / Deposit Value	Contract Management	Should	K14
Promise-to-Pay Date	CRM / Collection	Should	K09
Billing Dispute Status	Billing / CRM	Should	K12
Vehicle Utilization	Fleet Management	Nice	K13
Telematics / GPS Data	Telematics	Nice	K13
External Credit Information	External Provider	Nice	Future risk assessment

11.4 Data Quality Requirements

ID	Requirement	Validation Method
DQ-01	Customer and contract identifiers must be consistent	Referential integrity checks
DQ-02	Invoices must contain valid due dates	Completeness and validity checks

ID	Requirement	Validation Method
DQ-03	Payments must be traceable to their corresponding invoices	Payment allocation reconciliation
DQ-04	Due date changes must be historically traceable	Version history validation
DQ-05	Exposure values must support period-over-period comparison	Historical snapshot verification
DQ-06	Alerts and actions must contain timestamps	Audit trail validation
DQ-07	Prediction inputs must have been available before prediction time	Data leakage prevention
DQ-08	Missing or insufficient data must be identifiable	Insufficient Data validation

Critical Data Dependency

Invoice-level payment allocation and historical due date information are essential.

Without these elements, K06, K07, and the evaluation of K21 may produce misleading results.

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12. KPI-to-Action Traceability

The following table connects the eight MVP KPIs and the mandatory precision guardrail to operational triggers, responsibilities, processes, and data.

KPI	Trigger	Owner → Action → SLA	Business Process	Required Data
K02	Overdue AR >20%	AR Manager → Review portfolio → 1 business day	BP04	Due Date, Outstanding AR
K06	On-Time Payment Rate <80%	Credit Team → Review customer → 24 hours	BP04	Invoice and Payment History

KPI	Trigger	Owner → Action → SLA	Business Process	Required Data
K07	Payment Delay Trend >5 days	Collection → Confirm payment status → 24 hours	BP04, BP06	Due Date, Settlement Date
K10	Exposure Growth >20%	Credit Team → Review exposure → 24 hours	BP02, BP04	Historical Exposure
K15	Monitoring Coverage <85%	IT → Investigate monitoring gaps → 24 hours	BP02, BP06	Contract Status, Assessment Date
K17	SLA Compliance <75%	Collection Manager → Escalate overdue tasks → 24 hours	BP05	Alert Time, Action Time
K21	Pre-Due Recall <40%	Risk Analytics → Evaluate detection rules → 5 business days	BP06	Alerts, Due Dates, Actual Outcomes
K24	Alert Lead Time <2 days	Risk Analytics → Review alert timing → 5 business days	BP03, BP06	Due Date, First Alert
K22	Alert Precision <30%	Risk Analytics → Calibrate rules → 5 business days	BP06	Predicted and Actual Labels

[Assumption] The triggers and response times are initial proposals.

K06, K07, and K10 primarily support customer-level risk identification. K02, K15, K17, K21, K22, and K24 primarily support portfolio-level or system-performance monitoring.

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13. Logical Solution Architecture

13.1 Proposed Technology Components

Component	Proposed Technology	Purpose
Data Sources	Existing ERP / Finance Systems	Transactional and contractual information
Data Integration	Scheduled ETL / APIs / File Import	Data extraction and validation
Data Storage	Google Cloud Storage / Database	Data persistence
Behavioral KPI Engine	Python-Based Processing	K06, K07, and K10 calculations
Risk Engine	Rule-Based Scoring	Risk classification
Application Backend	Google Cloud Run	Application services
AI Assistance	Gemini AI	Risk explanations and suggested actions
Alert Management	Application / Notification Service	Alert generation and tracking
Dashboard	Web Application	Risk and KPI monitoring

[Assumption] These are proposed technology options, not a finalized deployment architecture. Actual services depend on budget, security requirements, and infrastructure availability.

13.2 Gemini AI Responsibilities

Gemini AI may be used to:

- Explain why a customer has been classified as high risk.
- Summarize changes in payment behavior.
- Generate suggested collection follow-up actions.
- Translate technical risk indicators into business-friendly explanations.

Gemini AI must not independently determine contractual actions or override the rule-based risk classification.

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14. Non-Functional Requirements

NFR ID	Requirement	Proposed Acceptance Criteria
NFR-01	Security	Role-based access control
NFR-02	Data Privacy	Customer information accessible only to authorized users
NFR-03	Auditability	Risk assessments, alerts, and actions are traceable
NFR-04	Reliability	Processing failures generate identifiable error states
NFR-05	Explainability	Risk classifications display their contributing indicators
NFR-06	Maintainability	Thresholds can be updated through configuration
NFR-07	Data Integrity	KPI calculations are traceable to source records
NFR-08	Human Oversight	High-impact decisions require authorized human approval
NFR-09	Cost Efficiency	MVP prioritizes existing data and lightweight infrastructure

[TBD] Quantitative requirements for availability, response time, recovery objectives, and user capacity remain undefined.

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15. Acceptance Criteria

15.1 Functional Acceptance Criteria

AC ID	Acceptance Criteria	Verification Method
AC-01	The system identifies eligible active contracts	Functional Testing

AC ID	Acceptance Criteria	Verification Method
AC-02	The system retrieves invoice and payment data	Integration Testing
AC-03	The system calculates K06 consistently	Calculation Testing
AC-04	The system calculates K07 using historical payment records	Calculation Testing
AC-05	The system calculates K10 using historical exposure	Calculation Testing
AC-06	The system generates risk classifications	Rule Testing
AC-07	The system generates pre-due alerts when conditions are met	Scenario Testing
AC-08	Alerts contain ownership and timestamp information	Functional Testing
AC-09	Collection actions can be recorded	User Acceptance Testing
AC-10	The system displays all eight MVP KPIs	Dashboard Testing
AC-11	Insufficient data produces the correct status	Negative Testing
AC-12	Gemini cannot independently modify risk scores	Integration Testing

15.2 Business Acceptance Criteria

Business KPI	Proposed Acceptance Target
K02	≥10% relative reduction in Overdue AR Ratio within 3–6 months
K15	Monitoring Coverage ≥95%
K17	SLA Compliance ≥90% within one business day
K21	Pre-Due Detection Recall ≥60%
K24	Alert Lead Time ≥5 days
K22	Alert Precision ≥50% [Assumption]

Targets for K06, K07, and K10 must be finalized after baseline assessment.

Functional acceptance can be demonstrated using controlled test data. However, predictive performance and business impact must be evaluated using representative historical or operational data.

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16. Risk Assessment and Mitigation

Risk ID	Risk Description	Potential Business Impact	Mitigation Strategy
R-01	Incomplete historical payment data	Inaccurate risk indicators	Data profiling and completeness validation
R-02	Missing invoice-level payment allocation	Incorrect payment delay calculations	Payment reconciliation before scoring
R-03	Excessive false-positive alerts	Increased Collection workload	Precision guardrail and threshold calibration
R-04	Data leakage during model evaluation	Overestimated detection performance	Historical cutoffs and temporal validation
R-05	Inaccurate Gemini recommendations	Potentially inappropriate business actions	Grounded explanations and human review
R-06	Alerts are not followed up	Early-warning benefits are not realized	Owner assignment and SLA monitoring
R-07	Contractual restrictions limit recommended actions	Recommendations cannot be implemented	Legal and Credit approval
R-08	Limited payment history	Unreliable risk scoring	Insufficient Data handling
R-09	Resistance to operational changes	Low system adoption	Pilot implementation and user training
R-10	Billing disputes affect payment behavior	Incorrect customer risk classification	Dispute flags and controlled exclusions

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17. MVP Scope and Implementation Roadmap

17.1 MVP Scope

Included in MVP	Excluded from MVP
Active contract monitoring	Full telematics integration
Invoice and payment history analysis	External credit bureau integration
K06, K07, and K10 calculations	Advanced machine learning
Rule-based risk classification	Automated repossession decisions
Pre-due alert generation	Automated credit approval
Alert assignment and action tracking	Automated payment restructuring
Dashboard for eight MVP KPIs	Full implementation of Phase 2 KPIs
K22 precision guardrail	Autonomous collection execution
Gemini-assisted risk explanations	AI-controlled financial decisions

17.2 Proposed Implementation Phases

Phase	Main Activities	Deliverables
Phase 1 — Discovery	Validate processes, data availability, and baselines	Approved business requirements
Phase 2 — Data Foundation	Integrate and validate source data	Validated AR dataset
Phase 3 — Risk Monitoring	Implement indicators and scoring	Functional risk engine
Phase 4 — Alert Management	Implement alerts, assignments, and action tracking	Functional alert workflow
Phase 5 — KPI Dashboard	Implement performance monitoring	MVP dashboard
Phase 6 — Pilot Evaluation	Conduct historical and operational evaluations	Pilot evaluation report

[TBD] Phase durations and implementation dates must be aligned with team capacity and competition deadlines.

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18. Measurement and Validation Methodology

18.1 Historical Evaluation

Detection effectiveness will be assessed by comparing information available before invoice due dates with actual payment outcomes.

[Assumption] Initial evaluation design:

Parameter	Proposed Definition
Prediction Window	D−14 to D−1
Prediction Cutoff	Before the invoice due date
Outcome Observation	Through D+7
Positive Outcome	Material overdue balance at D+7

Parameter	Proposed Definition
Materiality Threshold	Greater than max(IDR 1 million, 5% of invoice amount)
Evaluation Metrics	Recall, Precision, Alert Lead Time

These definitions require approval from Finance, Credit, and Collection stakeholders.

18.2 Evaluation Controls

The evaluation methodology must ensure that:

1. No information generated after the prediction cutoff is used as a prediction input.
2. Invoices without sufficiently mature outcomes are excluded from final evaluation.
3. Repeated alerts for the same invoice do not inflate detection counts.
4. Due date modifications and payment restructuring are historically recorded.
5. Preventive interventions are considered when interpreting detection outcomes.

18.3 Business Impact Evaluation

A reduction in Overdue AR Ratio after implementation does not automatically establish that the early-warning system caused the improvement.

Business impact evaluation should consider:

- Pre-implementation versus post-implementation performance.
- Changes in customer portfolio composition.
- Changes in credit sales volume.
- Changes in collection policies.
- A comparison or control group, where feasible.

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19. Open Issues and Required Decisions

ID	Open Issue	Required Decision	Proposed Owner
OI-01	Overdue AR baseline is unavailable	Confirm baseline and measurement period	Finance

ID	Open Issue	Required Decision	Proposed Owner
OI-02	K06, K07, and K10 targets remain undefined	Establish targets using historical data	Credit / Finance
OI-03	Eligible contract definition is not finalized	Define inclusion and exclusion criteria	Contract Management
OI-04	Payment allocation availability is unverified	Confirm data structure	Finance / IT
OI-05	Historical due date tracking is unverified	Confirm versioning capabilities	Billing / IT
OI-06	Risk scoring has not been calibrated	Validate weights and thresholds	Credit / Risk
OI-07	Collection SLA has not been formally approved	Confirm operational SLA	Collection Manager
OI-08	Six functional capabilities lack dedicated FR identifiers	Formalize additional requirements	Business Analyst
OI-09	ERP integration method is not confirmed	Select integration approach	IT
OI-10	Google Cloud operating costs are not estimated	Prepare cost estimate	Technical Lead
OI-11	Pilot evaluation methodology is not approved	Confirm evaluation design	Project Sponsor
OI-12	Gemini data-handling policies are undefined	Conduct security and privacy review	IT / Compliance

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20. Requirements Traceability Matrix

Business Objective	KPI	Functional Requirements	Business Process	Expected Outcome
BO-01	K02	FRI-05, FRI-06, FRO-05	BP04	Overdue AR is measured and reduced

Business Objective	KPI	Functional Requirements	Business Process	Expected Outcome
BO-02	K06	FRI-01, FRI-02, FRO-01, FRO-02	BP04, BP06	Customer payment behavior is monitored
BO-03	K07	FRI-03, FRI-04, FRO-03, FRO-04	BP04, BP06	Payment deterioration is detected
BO-04	K10	FRI-07, FRI-08, FRO-07, FRO-08	BP02, BP04	Financial exposure risk is monitored
BO-05	K15	FRI-17, FRI-18, FRI-19, FRO-15, FRO-16	BP02, BP06	Eligible contracts are monitored
BO-06	K17	FRI-09, FRI-10, FRI-11, FRO-09, FRO-10	BP05	Risk alerts receive timely responses
BO-07	K21	FRI-12, FRI-13, FRI-14, FRO-11	BP04, BP06	Pre-due detection effectiveness is measured
BO-08	K24	FRI-15, FRI-16, FRO-13	BP03, BP06	Alerts provide sufficient intervention time

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21. Document Approval

Role	Name	Approval Status	Date
Business Analyst	TBD	Pending	TBD
Finance / AR Representative	TBD	Pending	TBD
Collection Representative	TBD	Pending	TBD
Credit / Risk Representative	TBD	Pending	TBD
IT / Technical Lead	TBD	Pending	TBD
Project Sponsor	TBD	Pending	TBD

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22. Document Readiness Assessment

Document Status: Draft v0.2 — Ready for Stakeholder Review, Not Yet Approved

This BRD includes the following components:

BRD Component	Status
Business Problem and Objectives	Completed — Draft
Management Decisions D01–D07	Completed — Draft
AS-IS and TO-BE Business Processes	Completed — Pending Validation
25-KPI Framework	Completed — Thresholds Require Validation
Eight MVP KPIs	Defined
K22 Precision Guardrail	Defined
Business Requirements	Completed — Draft
Functional Requirements	Partially Complete — Six Gaps Identified
Data Requirements	Completed — Pending Source Verification
Business Rules	Completed — Draft
Risk Scoring Methodology	Proposed — Pending Calibration
Solution Architecture	Proposed — Pending Technical Review
Acceptance Criteria	Completed — Draft
Risk Assessment	Completed — Draft
Implementation Roadmap	Proposed
Requirements Traceability Matrix	Completed — Draft

Three Priorities Before Final Approval

1. Validate the company's actual data environment. Confirm the availability and structure of invoice records, payment allocations, outstanding receivables, contract information, and historical due dates.

2. Resolve the six functional requirement gaps. Daily assessment, risk scoring, risk classification, alert generation, action recording, and insufficient-data handling require dedicated and testable requirements.

3. Finalize KPI targets and the evaluation methodology. Particular attention should be given to the Overdue AR baseline, K06/K07/K10 targets, pre-due detection outcome definitions, and the method used to measure business impact.

Final Recommendation

For the **AI Builder Cup**, the proposed MVP should remain focused on a single measurable business objective:

> **Enable proactive identification and management of customer payment risks before invoices become overdue, using existing financial data and a cost-efficient AI-assisted monitoring solution.**

The eight priority KPIs establish an end-to-end measurement framework:

Monitoring Coverage → Payment Risk Identification → Pre-Due Detection → Early Alert → Collection Response → Overdue AR Reduction

This approach keeps the solution focused on a specific business problem while providing a traceable connection between management decisions, business requirements, operational processes, data, and measurable outcomes.

End of Business Requirements Document — Version 0.2